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Roll no.

Degree: B. Tech. Semester: 2nd sem (Improvement) — Course work
MID-SEMESTER EXAMINATION, February, 2025

Course Title: Electronic Devices and Circuits

Course Code: EAEPC02

Duration: 1:30 Hours

Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	A germanium diode carries a current of 10mA when it is forward biased with 0.2 V at 27 C Find the bias voltage required to get a current of 100 mA.	1.5	CO1
1b	Draw V-I characteristics of Semiconductor Diode and explain its functioning.	1.5	CO1
2a	Write difference between Zener and Schottky Diode.	1.5	CO1
2b	Write short notes on Varactor diode.	1.5	CO1
3a	What value of series resistance is required when three 10 W, 10 V , 1000 mA Zener diodes are connected in series to obtain a 30 V regulated output from a 45 V dc power source.	1.5	CO2
3b	Comment on Physical structure of Zener Diode	1.5	CO2
4a	Explain the Construction and working operation of BJT using suitable diagrams.	1.5	CO2
4b	Draw input and output characteristics of CE NPN BJT.	1.5	CO2
5a	An npn silicon transistor has $V_{cc}=6$ V and the collector load $R_c=2.5$ kohm. Find the maximum collector current that can be allowed during the application of signal for faithful amplification.	1.5	CO2
5b	Explain the concept of load line and Operating point for a NPN BJT using suitable diagram.	1.5	CO2

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MID Semester of VIIth Sem – B.Tech. (ECAM)

MID-SEMESTER EXAMINATION, FEBRUARY 2025

Course Code : EAEPE63

Course Title : Communications Beyond 5G

Time- 1.5 Hours

Max. Marks-15.

Note: - Attempt all Questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer. Use of Calculator is allowed.

Q.No.	Question	Marks	CO
1 (a)	Describe Challenges for stratification of 5G evolution and B5G.	1.5	CO1
1 (b)	Define 5G evolution baseline	1.5	CO1
2 (a)	Describe B5G technology enablers	1.5	CO1
2 (b)	Describe what are the vision and goals for Communications in 6G	1.5	CO1
3 (a)	Describe the Future of Radio Access	1.5	CO1
3 (b)	Describe Terahertz Communication and the Opportunities of Wireless Technology Beyond 5G	1.5	CO2
4 (a)	Describe tactile internet (TI) in the beyond 5G era	1.5	CO2
4 (b)	Describe security and privacy for TI applications	1.5	CO2
5 (a)	Describe sub-millimeter wave/THz communication	1.5	CO2
5 (b)	Describe the Implementation Challenges and Opportunities in Beyond-5G and 6G Communication	1.5	CO2

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SIXTH SEMESTER B. Tech.
MID-SEMESTER EXAMINATION, February 2025

Course Code- EAEPE20

Course Title- Deep Learning

Time- 1 Hours

Max. Marks- 25

Note: - Attempt all questions. Missing data/ information if any, may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	What is Deep Learning? Give pictorial representation of Neural network when compared to human brain.	3	CO 1
1b	Describe how different parts of an Artificial Intelligence system relate to each other within different AI disciplines with diagram.	2	CO1
2a	Describe different types of learning algorithms in machine learning. Write difference between Deep learning and machine learning.	3	CO1
2b	Give difference between parameters and hyperparameters.	2	CO1
3a	When do you use backpropagation in neural network?	2	CO1
3b	Explain Gradient Decent algorithm used in backpropagation.	3	CO2
4a	What is the difference in SGD and Minibatch explain with the help of algorithms and examples?	3	CO2
4b	Why momentum methods are used in optimizers? Name some methods which use momentum method of optimizing.	2	CO2
5a	For given 4 choices of input, design the weights matrix for the condition that AND gate is satisfied for all the input points.	3	CO2
5b	How do you update weight and biases in case of a parabolic equation using Gradient descent algorithm. Evaluate numerically.	2	CO2

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VITH SEMESTER-BTECH-CIOT
MID SEMESTER EXAMINATION, Feb 2025

Course Code: CICPE06

Course Title: Computer Graphics & Multimedia

Time: 1.30 Hours

Max Marks: 15

Q. No	All questions are compulsory.	Marks	CO
1a	Define multimedia and list its key components.	2	CO1
1b	Describe Cathode Ray Tube (CRT) and list its main components.	1	CO1
2a	Explain the role of a rendering engine in computer graphics.	2	CO1
2b	Compare and contrast lossy and lossless compression techniques, providing a relevant example for each.	1	CO1
3a	Explain an algorithm for drawing line in computer graphics.	1.5	CO2
3b	How are images stored in Raster Graphics?	1.5	CO2
4a	An image has a resolution of 1920 x 1080 pixels. If each pixel requires 3 bytes to store its color information, calculate the total size of the image in megabytes (MB).	2	CO2
4b	What is anti-aliasing?	1	CO1
5a	Explain the key difference between Raster Graphics and Vector Graphics, highlighting their respective advantages and disadvantages.	2	CO2
5b	Write a short explanation of the concept of a 'bitmap' in computer Graphics, and mention its key characteristics	1	CO1